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Characterization of blood samples using image processing techniques

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ABSTRACT

This paper presents a methodology to determine human blood types in an emergency situation as well as a reliability study of the methodology proposed. The plate test is employed to determine blood types, allowing the macroscopic results observation. A CCD camera captures an image of the plate test results that will be processed through image processing techniques available in the IMAQ Vision software from National Instruments. The techniques used in this work are able to determine the occurrence of agglutination allowing the determination of blood types with an algorithm of classification. The reliability study based on statistical tests determines the level of confidence of the approach. The equipment developed is able to automatically perform the plate test and determine blood types.

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1. Introduction

Determining the blood type of an individual is an essential procedure before administering a blood transfusion, avoiding damage and blood incompatibilities. Thus, the preceding blood test, even in case of medical emergency, is of up most importance, allowing the administration of the blood type of the receiver in the first unit of blood transfusion. However, although two of the manual tests used in laboratory, the plate test and tube test, allow the analysis in a small time interval, which is suitable for emergency situations, they require travel to the lab and the intervention of a technician, thereby increasing the time spent and the possibility of human error. Given that human errors in the procedure of manual tests, in reading or interpreting the results, can lead to fatal consequences for patients, being one of the significant causes of fatal blood transfusions is extremely important to automate the testing, the reading and the interpretation of results [1–10].

In this sense, various systems were developed, Technicon Auto Analyzer [11–13], Technicon Auto Analyzer II [14,15], Groupamatic [12,16], Auto-Grouper [12] Olympus PK 7200 [12,17–20] Immucor Galileo [21–24], Ortho AutoVue® Innova System [25], Tango® Automated Blood Bank [26] and Techno TwinStation [27]. But, none of the systems currently available allow obtaining results on useful time so they can be used in emergency situations [25]. This paper presents an innovative system to automatically perform tests to determine blood type, eliminating human error and in a short interval of time, so they can be used in emergency situations.

This system is based on the plate test for determining blood types and in software developed, using image processing techniques. The plate test consist of the mixture of one drop of blood, and one drop of each reagent, anti-A and anti-B, anti-AB and anti-D, being the result interpreted according to the occurrence or not of agglutination. The agglutination means that a reaction has occurred between the antibody and the cells indicating the presence of the appropriate antigen. The combination of the occurrence of agglutination, or non-occurrence, determines the blood type of the individual [28]. Thus, the developed software allows, through image techniques, to detect the occurrence of agglutination and consequently the blood type of an individual.

2. Image processing techniques

The result of the plate test is captured by a CCD camera (Sony Cyber-shot DSC-S750) consisting in a color image composed of four samples of blood and reagent. This image will suffer a series of transformations that change, as it is processed by image processing techniques developed with the IMAQ Vision software from National Instruments [29]. However, since the original image also needs to be used again at a later stage of processing, it must remain available and unchanged. The following sequences of image processing techniques are used:

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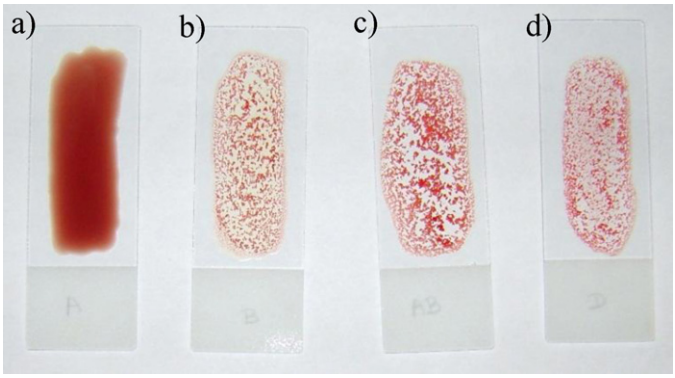

Fig. 1. Original image captured by the CCD camera (a) reagent anti-A, (b) reagent anti-B, (c) reagent anti-AB, and (d) reagent anti-D.

2.1. Image buffer: add copy

This feature allows saving a copy of the original color image captured by the CCD camera after performing the test procedure for determining the blood type. It is the first function to be used as it is necessary to keep the original image intact, i.e. before undergoing transformations carried out by other image processing functions. The image is stored in Buffer # 1 and is available for later use [29–31].

The result of applying this function is shown in Fig. 1, for the four reagents, anti-A and anti-B, anti-AB and anti-D.

2.2. Color plane extraction: RGB green plane

This function is used to extracting green plans in a RGB image, allowing transforming the original color image composed of 32 bit in a grayscale image of 8 bit. This step is essential in order to use functions like threshold that can only be applied to binary images or images with 8 bit, 16 bit or float [29–31].

The image obtained by applying this function to the image of Fig. 1 is shown in Fig. 2.

2.3. Auto threshold: clustering

This function applies a threshold based on a statistical technique called clustering and it is used to separate the blood samples and reagent into two regions, designated "particle region" and "background region". This process is accomplished by changing all pixels that lie between the range of values established (designated threshold range) changing all other image pixels to 0 [29–31]. Thus, to isolate the particles of interest and convert the gray scale image (pixel values between 0 and 255) into a binary image (pixel values of 1 and 0).

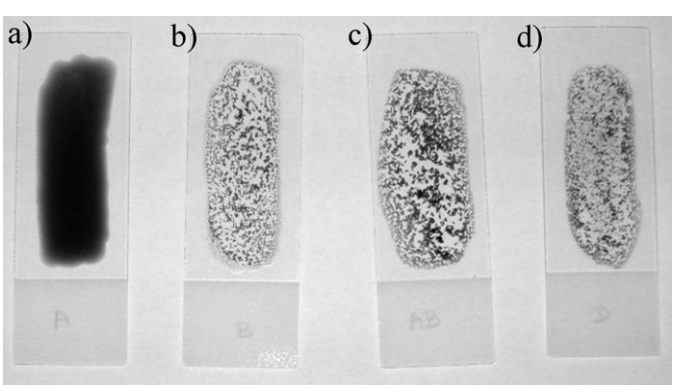

Fig. 2. Image obtained by applying color plane extraction: RGB green plane function in the original image of Fig. 1.

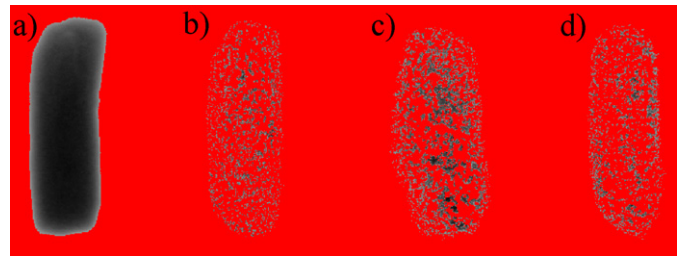

Fig. 3. Image obtained by applying the auto threshold: clustering in the image of Fig. 2.

This function is an automatic threshold, as it is not necessary to specify the minimum and maximum light intensity. To draw the threshold, it is used the image histogram [29–31].

The image obtained with the application of this function is shown in Fig. 3.

Now that the particles of interest are separated from the background of the image, it is necessary to isolate the four particle image and eliminate any unwanted particles. So, there were used the following three functions, local threshold: niblack, advanced morphology: fill holes and advanced morphology: remove small objects.

2.4. Local threshold: niblack

In this function the threshold value for each pixel is calculated based on statistics from adjacent pixels. It uses a kernel, by default, of 32 width and 32 height, with a deviation factor which by default is 0.20 [29–31]. This function is extremely important to isolate the particles to be analyzed.

The result of applying this function on the image of Fig. 3 can be seen in Fig. 4.

2.5. Advanced morphology: fill holes

This function fills all the holes that are present inside the particles with a pixel of value 1. The binary image resulting from the application of this function contains particles intact, without holes, corresponding to samples of blood and reagent [29–31]. The result of applying this function is presented in Fig. 5.

2.6. Advanced morphology: remove small objects

This function uses a low pass filter to remove small particles and the possible noise in the binary image resulting from the previous function.

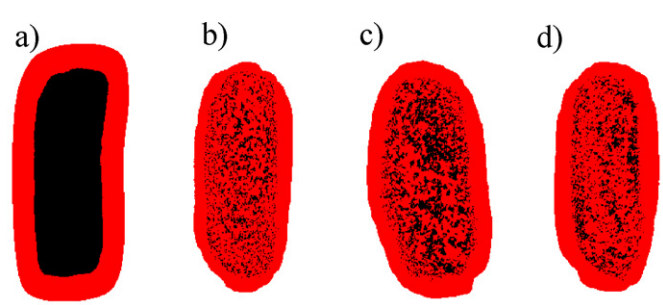

Fig. 4. Image obtained by applying the local threshold: niblack function under the image of Fig. 3.

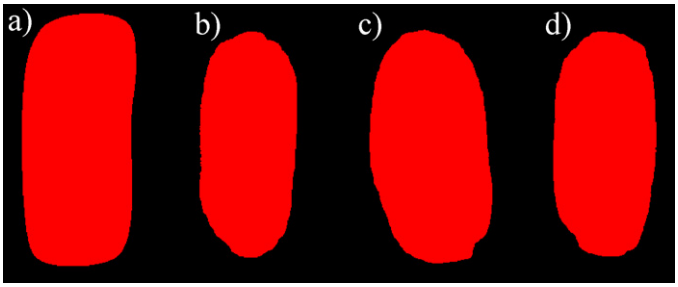

Fig. 5. Image resulting from the application of advanced function morphology: fill holes function in the image of Fig. 4.

The particles are removed according to their widths, as defined by a specific parameter, filter size. Thus, for a given filter size *N*, the low pass filter removes the particles whose widths are less than or equal to (N−1) pixels. These particles disappear after (N−1)/2 erosions [29–31].

The function removes particles that are not relevant to the analysis, leaving the rest of the particles (larger particles) untouchable. The particles that are removed from the step of erosion particles are assumed to be noise. It is mainly used to remove particles that may interfere with the analysis of the image, like little drops of blood or reagent at the background of the image [29–31]. The result of this function is presented in Fig. 6. In this case, as there are only very small particles in Fig. 5, which are difficult to see at naked eye, the application of this function is not noticeable in Fig. 6. However, when there are small drops of liquid (blood or reagent) present in the image, it becomes crucial to use this function to remove the unwanted particles.

The functions presented at this stage are essential for the automation of the process of determining the blood type.

2.7. Particle analysis

A particle is a continuous group of nonzero values in an image. The particles can be characterized by measures related to their attributes such as location of particles, area, and shape, among others. Therefore, this step is used to analyze the properties of particles in the image, considered as four particles corresponding to each of the mixtures of blood and reagent.

These particles can be analyzed using various properties, being the center of mass highly important in this work.

The result of this function is a table that contains all the selected properties and their values for each of the particles. The values of the center of mass *X* (coordinate *X* of the center of mass) and the center of mass *Y* (coordinate *Y* of the center of mass) for each particle are shown in Table 1.

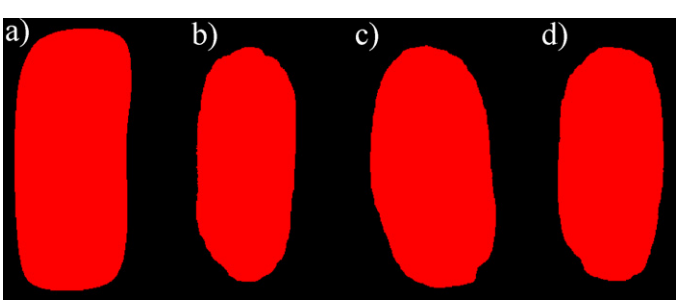

Fig. 6. Image resulting from the application of advanced function morphology: remove small objects in the image of Fig. 5.

Table 1
Results of application of particle analysis function on image of Fig. 6.

Particle	Center of mass *X*	Center of mass *Y*
1	338.48	537.65
2	1163.82	557.58
3	735.68	545.93
4	1570.67	545.81

2.8. Image buffer: retrieve buffer # 1

This function retrieves a copy of the original color image stored in buffer # 1, so it can be used in the next function. The original image consists of four samples of blood and reagents that will be processed and analyzed to detect the occurrence of agglutination [29–31]. The original image, as previously referred, is shown in Fig. 1.

2.9. Color plane extraction: HSL luminance plane

This function is used to extract the planes of a color image light obtained with the previous function [29–31] and allows transforming the original image constituted of 32 bit in a grayscale image constituted of 8 bit. Thus, it makes possible the use of the Quantify function that can only be applied to images of binary type, or images with 8-bit and 16 bit float, or grayscale images.

The image obtained by applying this function to the image of Fig. 1 is shown in Fig. 7.

2.10. Set coordinate system

This function defines a coordinate system based on the stage of particle analysis. The function of particle analysis obtains the coordinates (center of mass *X* and center of mass *Y*) needed to calculate the center of mass used in this function. This function is essential because it is based on the coordinate system, which is selected to analyze the region, depending on the center of mass calculated in the step of particle analysis. Thus, the selection process in the region of interest analysis is automatic and depends on the center of mass of each particle. This step is repeated for each of the particles to be analyzed. In this case it will be repeated four times, one for each sample of blood and reagent [29–31].

The image obtained by applying this function to each particle image is shown in Fig. 8.

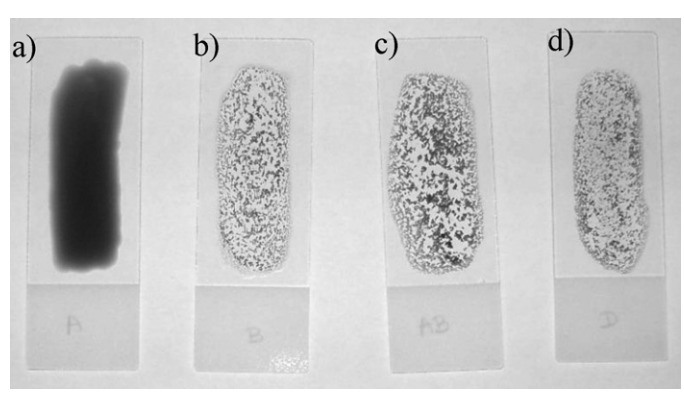

Fig. 7. Image obtained by applying the color plane extraction: HSL luminance plane function to the original image of Fig. 1.

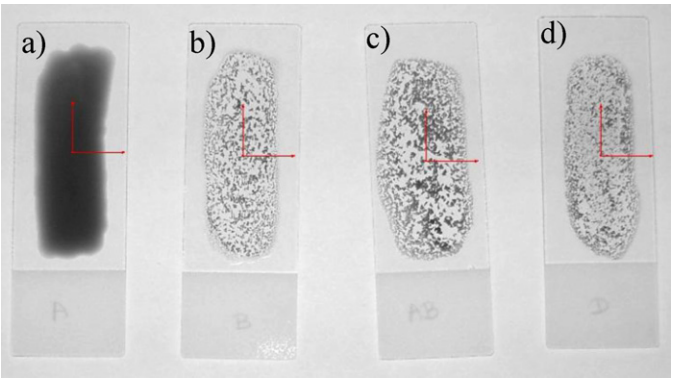

Fig. 8. Image resulting of the application of set coordinate system functions to each of the particles in the image.

Table 2
Results of application of the quantify function in Fig. 9.

Fig.	Area	Mean value	Standard deviation	Minimal value	Maximal value
9(a)	0.5	57.6	5.2	43.0	83.0
9(b)	0.4	193.5	43.7	63.0	250.0
9(c)	0.5	176.9	53.4	46.0	248.0
9(d)	0.4	182.9	25.5	61.0	234.0

2.11. Quantify

This function can measure the intensity of pixels in the region of interest selected. This step uses the options, Reposition Region of Interest and Reference Coordinate System. The first option, Reposition Region of Interest, dynamically restores the region of interest based on the coordinate system previously defined. The second option, Reference Coordinate System, allows to select the coordinate system associated to the region of interest. The analysis of each sample is automatically performed.

The result of this function is a table that contains the area (percentage of surface examined for the full image), the mean (average value of pixels), standard deviation (standard deviation of pixels), the minimum and maximum value of pixels intensity.

Using this feature, it allows identifying the occurrence of agglutination in the blood sample and reagent based on the value of standard deviation. As in the previous function, this step is repeated for each of the particles to be analyzed [29–31].

The image obtained by applying this function is shown in Fig. 9 and the resulting table is shown in Table 2.

The value of standard deviation is essential to detect the occurrence of agglutination. In samples where no agglutination occurs, the values of standard deviation do not exceed 16, taken normally,

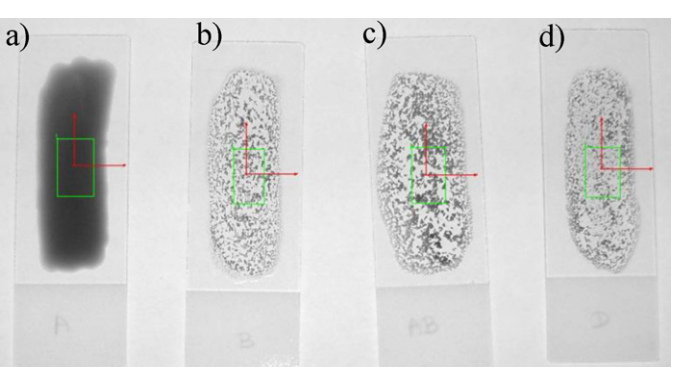

Fig. 9. Image resulting from the application of the quantify function.

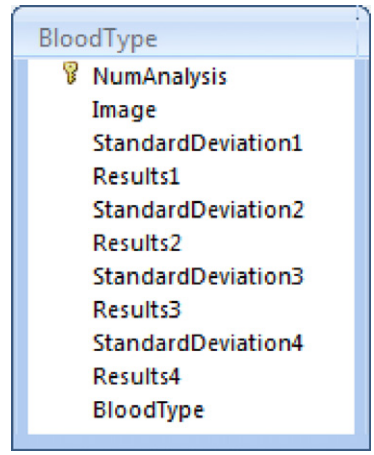

Fig. 10. Base table where the information is entered.

values between 0 and 10. Moreover, samples where agglutination occurs show standard deviation values greater than or equal to 16, showing even high standard deviation values between 20 and 70. Thus, it was established as a threshold value for classification, a standard deviation of 16;samples with a standard deviation value below 16 are classified as samples with no agglutination and samples with a standard deviation value greater than or equal to 16 are classified as samples in which agglutination occurred.

3. Database

To store the information from the analysis of detection of agglutination performed through the techniques of image processing and the result of the classification algorithm developed to determine blood type, it was designed and implemented a database. This

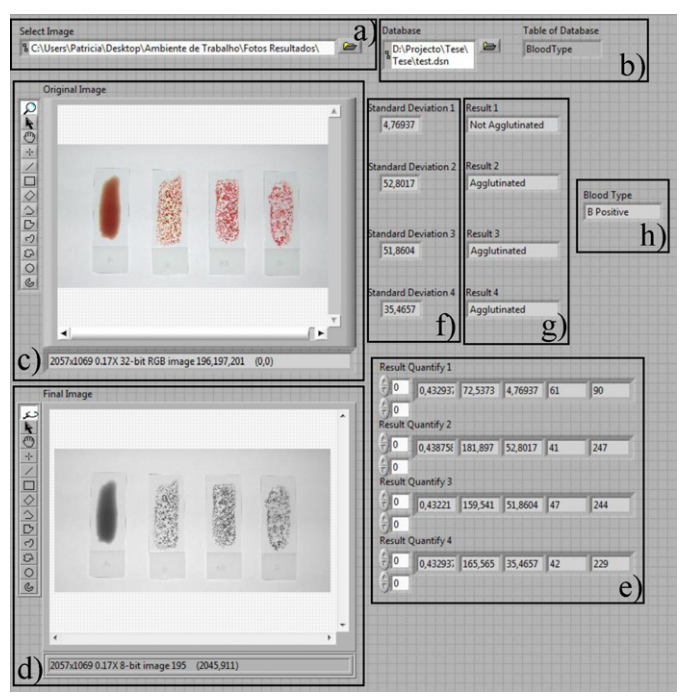

Fig. 11. Developed software. (a) Selection of the image, (b) selection of file database dsn and entering the name of the table of database, (c) original image, (d) final image obtained with the image processing techniques, (e) results of the quantify function, (f) values of standard deviation of each sample, (g) results based on the standard deviation of each sample, and (h) result of the classification algorithm.

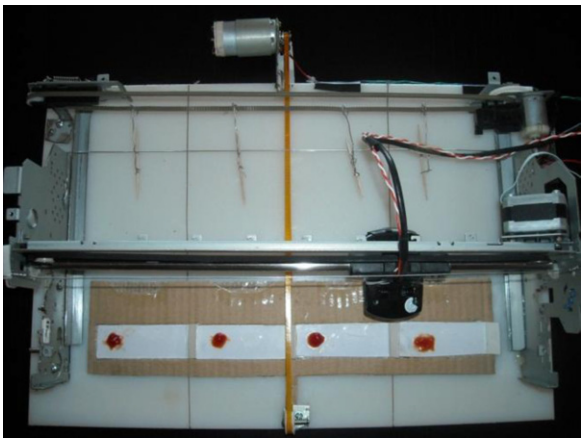

Fig. 12. System developed.

section presents the table of the database developed with Microsoft Office Access 2007, where information will be stored, Fig. 10.

This table, Fig. 10, stores the image captured and used in image processing (each image contains four samples of blood and reagents), the standard deviation calculated in each sample image, the occurrence or not of agglutination and the blood type.

4. Software developed

The software developed allows, through an image captured by a CCD camera (in this case Sony Cyber-shot DSC-S750), to detect the occurrence of agglutination through image processing techniques developed and presented in Section 2, to determine the blood type, through the classification algorithm developed and to store the information in the database constructed and presented in Section 3.

The software was developed in LabView from National Instruments [30]. The front panel is shown in Fig. 11.

This software allows to select the image to be used, Fig. 11(a), the database to establish a connection and the table from the database where information is added to the analysis, Fig. 11(b), displays the original image that is being used, Fig. 11(c) as well as the final image obtained with the techniques of image processing, Fig. 11(d), shows the results of the quantify function, Fig. 11(e), the results of the standard deviation, Fig. 11(f), the results based on standard deviation values calculated Fig. 11(g), and the blood type in question, Fig. 11(h).

5. System developed

The developed system, which automatically determines the blood type of an individual, using as reference the plate test [28], is shown in Fig. 12.

The system requires that the blood samples and the reagents are manually placed, in plates, by the user. It is placed on the first plate the reagent anti-A, in the second plate the reagent anti-B, in the third plate the reagent anti-AB and in the fourth plate the reagent anti-D. Thereafter, the system moves the plates for the mixing area, where the blood and the reagents in each plate, are mixed. This mixture is performed with a DC motor and without contamination between the four samples. Upon completion of the mixing, blades are moved to the image capture area, where a motor drives the webcam Glossy, 5 Mega pixels, along the samples, capturing an image of each of the plates. These images are stored for later analysis. The system is controlled by a microcontroller.

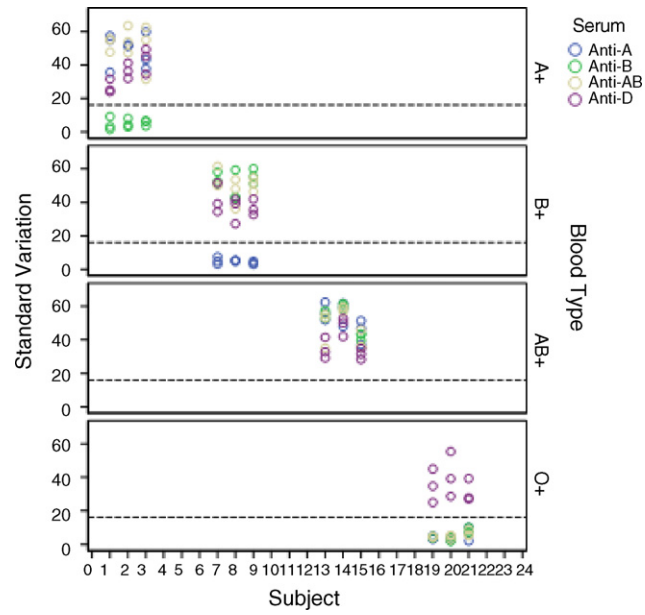

Fig. 13. Standard deviation values obtained by the image processing techniques, for all individuals Rh Positive (D Positive).

6. Statistical analysis

This section presents the statistical analysis used to prove that the boundary between the occurrence of agglutination and non-agglutination is reliable. Namely, there are no values of agglutinated samples below the limit (16) and there are no values upper the limit (16) for samples that are not agglutinated.

This statistical analysis allows determining the reliability of the methodology used as well as the limit set on the basis of the statistical tests. To this end, it was used blood of 24 individuals and determining the blood type of each one, using the software developed in this work. For each individual, the procedure for determining the blood type was repeated three times in order to verify whether the results were similar. There were obtained a total of 288 results for analysis. It is noteworthy that the blood type of the tested individuals was previously determined by the technique of determining the blood type currently used in laboratories, Cards-ID [32] in order to verify the validity of the technique developed in this work.

The results of statistical analysis were obtained using SPSS Software (Version 17.0) [33], being the standard deviation value used to analyze the reliability of the technique. The statistical analysis was based on descriptive data analysis. The study of the variability for each blood type results is shown in Figs. 13 and 14.

The study of variability was performed, initially for individuals' Rh Positive and at a second stage, for individuals Rh Negative. The dashed line represents the limit of standard deviation, 16, set out to determine the occurrence of agglutination, or non-occurrence. For all samples, there were no standard deviation values that could lead to wrong decisions. For example, in case of blood type A+ (A Positive), agglutination was not observed for the reagent anti-B in all tests (Fig. 13, green). For this case, all standard deviation values obtained are below the threshold. For the case of blood type O-(O Negative), agglutination does not occur in all reagents and in all tests, all standard deviation values obtained are below 16 (Fig. 14). On the other hand, for blood type AB+ (AB Positive), all values are greater than 16 (Fig. 13).

For each test the agglutination was detected always getting standard deviation values greater than or equal to 16. The same was

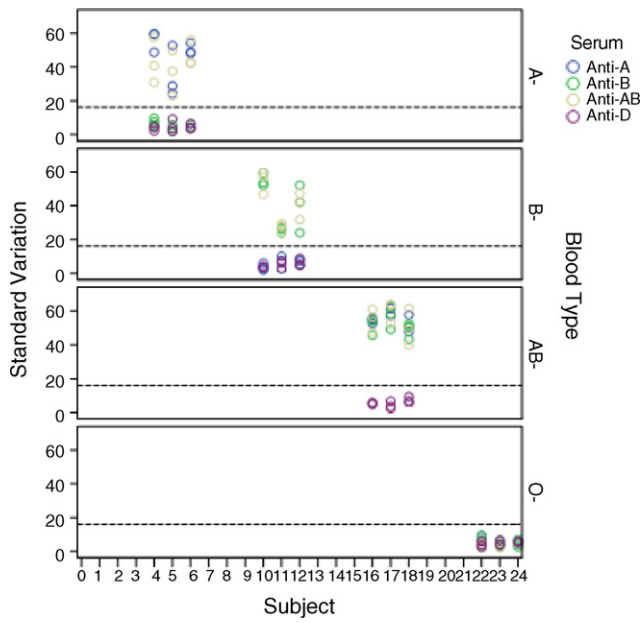

Fig. 14. Standard deviation values obtained by the image processing techniques, for all individuals Rh Negative (D Negative).

Table 3
Results of application of quantify function on image of Fig. 16.

Fig.	Area	Mean value	Standard deviation	Minimal value	Maximal value
16(a)	0.5	144.3	57.8	20.0	238.0
16(b)	0.5	83.4	3.4	64.0	95.0
16(c)	0.4	147.4	62.1	32.0	240.0
16(d)	0.4	139.3	49.5	34.0	222.0

true for the absence of agglutination, obtaining values of standard deviation always below 16.

7. Results

This section presents the experimental results obtained. Each blood sample group tested in the prototype, and therefore by the image processing techniques, was previously determined by two conventional methods, ID-Card test and test lamina.

Observing Fig. 15, agglutination occurred in samples (a), (c) and (d); in sample (b), there was no agglutination. Through the standard deviation values of Table 3, this is also confirmed as samples (a), (c), (d), present values of standard deviation of 57.8, 62.1 and 49.5, respectively (all above 16), sample (b), obtained 3.4 (below 16).

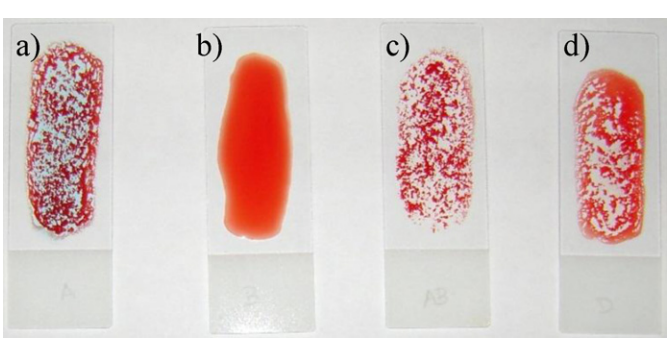

Fig. 15. The original image of A Positive (a) reagent anti-A, (b) reagent anti-B, (c) reagent anti-AB, and (d) reagent anti-D.

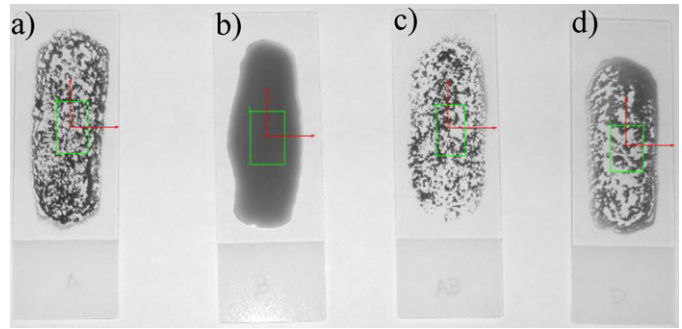

Fig. 16. Image resulting from application of image processing techniques presented in Section 2 to the image of Fig. 15.

Having verified the occurrence of agglutination in anti-A reagents, anti-AB and anti-D, it is confirmed the presence of A antigens and Rh antigens in the blood sample analyzed. Thus, it can be concluded that the blood type in question is A Positive.

8. Conclusions and future work

The methodology used in this work (previously validated by conventional methods) is capable to determinate blood types safely and in a short time (approximately 2 min) allowing it to be used in emergency situations. As demonstrated, the techniques applied are able to detect the agglutination (using a standard deviation value of 16 as threshold), and consequently blood types. The equipment developed is able to determine blood types using the plate test and to automatically perform the analysis, with exception to the introducing of blood and reagents that at this stage of development is performed manually. For future work, it is intended to evolve the system, reducing its size, automate all the processes and incorporate GSM technology, among other features.

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